



CANDIDATE
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0620/62

October/November 2023

1 hour

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].
- Notes for use in qualitative analysis are provided in the question paper.

This document has **16** pages. Any blank pages are indicated.

- 1 Some seashells contain a mixture of the insoluble compounds calcium carbonate and silicon(IV) oxide only.

Calcium carbonate reacts with dilute hydrochloric acid to form the soluble salt calcium chloride. Silicon(IV) oxide does **not** react with or dissolve in dilute hydrochloric acid.

A student wants to find the percentage of silicon(IV) oxide in a seashell. The first four steps of the method the student uses are shown.

- step 1** The student grinds the seashell to form a powder.
- step 2** The student finds the mass of the powdered seashell.
- step 3** The student adds the powdered seashell to an excess of dilute hydrochloric acid and heats while stirring with a glass rod as shown in Fig. 1.1.
- step 4** The student filters the mixture as shown in Fig. 1.2.

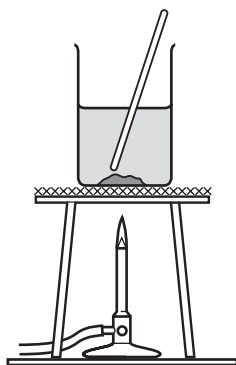


Fig. 1.1

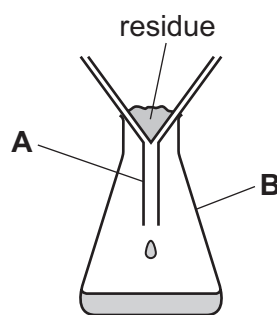


Fig. 1.2

- (a) Name the apparatus used to grind the seashell to form a powder in **step 1**.

..... [1]

- (b) Explain why it is important that the dilute hydrochloric acid is in excess in **step 3**.

.....
..... [1]

- (c) Name the items of apparatus labelled **A** and **B** in Fig. 1.2.

A

B

[2]

(d) The residue obtained in **step 4** is not pure.

- (i) Identify **one** substance, other than water, that is in the residue and prevents it from being pure.

..... [1]

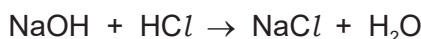
- (ii) The student washes the residue.

Describe **two** additional steps the student must now take to find the percentage of silicon(IV) oxide in the seashell.

.....
.....
.....
..... [2]

[Total: 7]

- 2 A student investigates the temperature change when aqueous sodium hydroxide neutralises dilute hydrochloric acid. The equation for the reaction is shown.



The student does six experiments.

Experiment 1

- Fill a burette with dilute hydrochloric acid.
- Run some of the dilute hydrochloric acid out of the burette so that the level of the dilute hydrochloric acid is on the burette scale.
- Fill a second burette with aqueous sodium hydroxide.
- Run some of the aqueous sodium hydroxide out of the burette so that the level of the aqueous sodium hydroxide is on the burette scale.
- Run 1.0 cm^3 of dilute hydrochloric acid from the burette into a boiling tube.
- Run 9.0 cm^3 of aqueous sodium hydroxide from the second burette into the same boiling tube.
- Stir the mixture with a thermometer and measure the highest temperature reached.
- Measure the pH of the mixture in the boiling tube.
- Rinse out the boiling tube with distilled water.

Experiment 2

- Run 2.0 cm^3 of dilute hydrochloric acid from the burette into the boiling tube.
- Run 8.0 cm^3 of aqueous sodium hydroxide from the second burette into the same boiling tube.
- Stir the mixture with a thermometer and measure the highest temperature reached.
- Measure the pH of the mixture in the boiling tube.
- Rinse out the boiling tube with distilled water.

Experiment 3

- Repeat Experiment 2 using 3.0 cm^3 of dilute hydrochloric acid and 7.0 cm^3 of aqueous sodium hydroxide.

Experiment 4

- Repeat Experiment 2 using 6.0 cm^3 of dilute hydrochloric acid and 4.0 cm^3 of aqueous sodium hydroxide.

Experiment 5

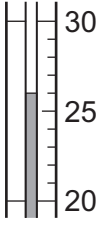
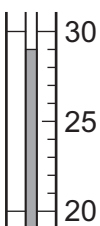
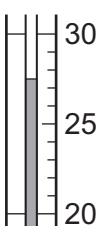
- Repeat Experiment 2 using 7.0 cm^3 of dilute hydrochloric acid and 3.0 cm^3 of aqueous sodium hydroxide.

Experiment 6

- Repeat Experiment 2 using 8.0 cm^3 of dilute hydrochloric acid and 2.0 cm^3 of aqueous sodium hydroxide.

(a) Use the description of the experiments and the thermometer diagrams to complete Table 2.1.

Table 2.1

experiment	volume of dilute hydrochloric acid / cm ³	volume of aqueous sodium hydroxide / cm ³	thermometer diagram when highest temperature reached	highest temperature reached / °C	pH
1	1.0				11
2	2.0				11
3	3.0				11
4	6.0				1
5	7.0				1
6	8.0				1

[4]

- (b) Add a suitable scale to the y-axis in Fig. 2.1. **Your scale should extend by 2°C above your highest temperature in Table 2.1.**

Plot your results from Experiments 1 to 6 on the grid.

Draw **two** straight lines through your points, one through the first three points and one through the last three points. Extend your straight lines so that they cross.

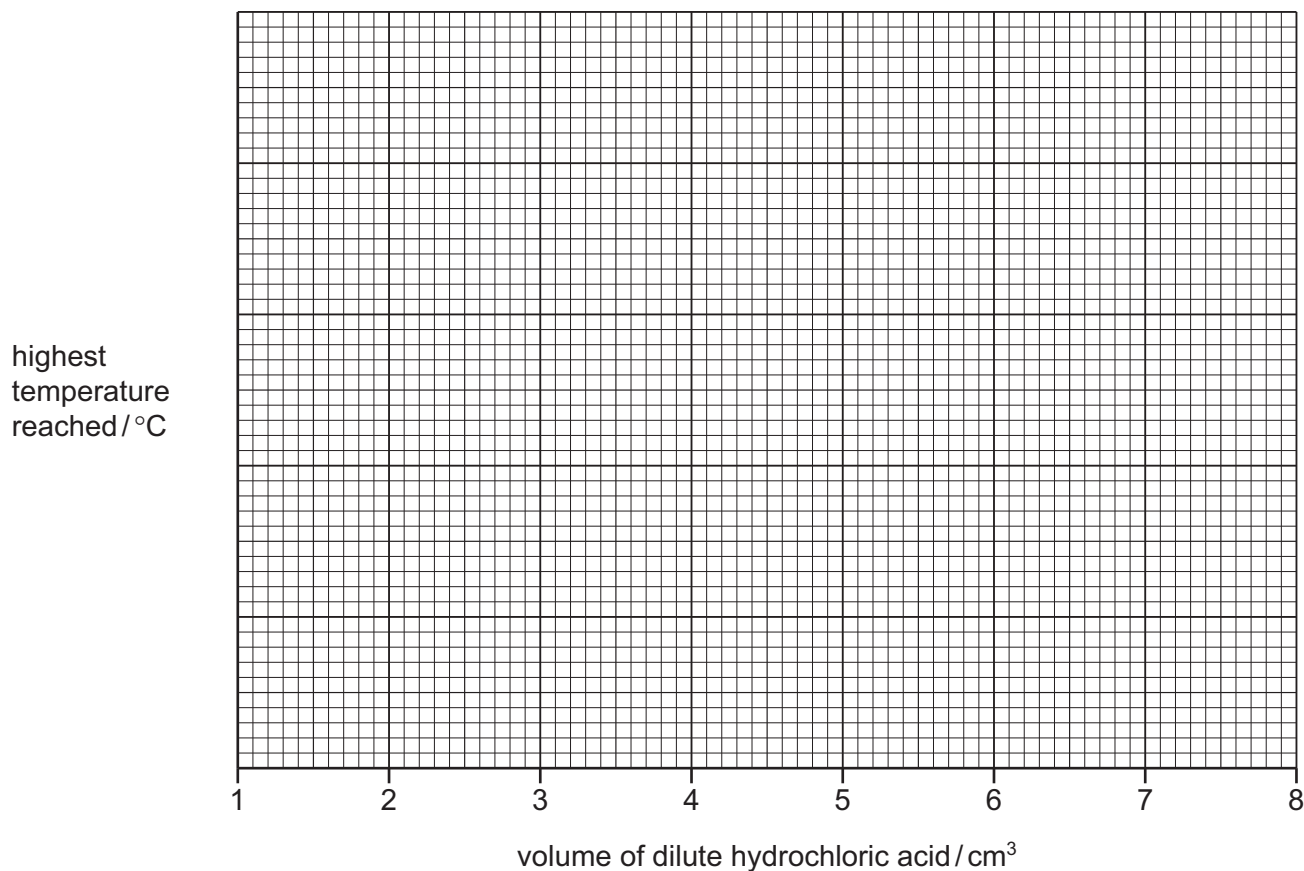


Fig. 2.1

[5]

- (c) The point on the graph where the two straight lines cross is where all of the aqueous sodium hydroxide reacts with all of the dilute hydrochloric acid to form a neutral solution.

- (i) **Use your graph** in Fig. 2.1 to deduce the volume of dilute hydrochloric acid and the volume of aqueous sodium hydroxide that react together to produce a neutral solution. Show your working **on Fig. 2.1**.

volume of dilute hydrochloric acid = cm³

volume of aqueous sodium hydroxide = cm³

[3]

- (ii) Predict the pH of the solution in the boiling tube when the volumes in (c)(i) are mixed together.

pH = [1]

- (iii) Deduce which solution, dilute hydrochloric acid or aqueous sodium hydroxide, is the most concentrated.

Use your answer to (c)(i) to explain why.

most concentrated solution

explanation

..... [1]

- (d) State how the pH and temperature recorded in each experiment would differ, if at all, if a polystyrene cup is used in place of the boiling tube.

Explain any differences.

pH

temperature

explanation

..... [3]

- (e) The volumes of the solutions used in these experiments were measured using a burette.

Explain why a volumetric pipette could **not** be used instead of a burette in this experiment.

..... [1]

[Total: 18]

- 3 A student tests two substances: solid **K** and solid **L**.

Tests on solid **K**

The student dissolves solid **K** in water to form solution **K**. The student divides solution **K** into four portions.

Table 3.1 shows the tests and the student's observations for solution **K**.

Table 3.1

tests	observations
test 1 To the first portion of solution K , add a few drops of aqueous ammonia.	white precipitate
test 2 To the second portion of solution K , add a few drops of acidified aqueous potassium manganate(VII).	pale purple solution
test 3 To the third portion of solution K , add 1 cm ³ of dilute nitric acid followed by a few drops of aqueous silver nitrate.	cream precipitate
test 4 To the fourth portion of solution K , add aqueous chlorine.	the solution becomes orange

- (a) (i) Identify **two** cations that **test 1** shows could be in solid **K**.

.....
 [2]

- (ii) Describe an additional test that could be carried out on solution **K** to confirm which of the two cations you have identified in (a)(i) is in solid **K**.

Explain how the test will show which of these two cations is in solid **K**.

test
 explanation
 [2]

- (b) Identify the anion in solid **K**.

..... [1]

Tests on solid L

Solid **L** is barium nitrate.

Complete the expected observations.

(c) The student carries out a flame test on solid **L**.

observations [1]

The student dissolves the remaining solid **L** in water to form solution **L**.

The student divides solution **L** into three portions.

(d) To the first portion of solution **L**, the student adds a piece of aluminium foil and 5 cm³ of aqueous sodium hydroxide and warms the mixture. The student tests for any gas produced.

observations
..... [1]

(e) To the second portion of solution **L**, the student adds 1 cm³ of dilute nitric acid and a few drops of aqueous silver nitrate.

observations
..... [1]

(f) To the third portion of solution **L**, the student adds 1 cm³ of dilute sulfuric acid.

observations
..... [1]

[Total: 9]

- 4** The solubility of solid sodium sulfate in water changes as the temperature of the water changes.

Plan an experiment to find out how the solubility of sodium sulfate in water changes with temperature.

You are provided with sodium sulfate, distilled water and common laboratory apparatus.

[6]

Notes for use in qualitative analysis

Tests for anions

anion	test	test result
carbonate, CO_3^{2-}	add dilute acid, then test for carbon dioxide gas	effervescence, carbon dioxide produced
chloride, Cl^- [in solution]	acidify with dilute nitric acid, then add aqueous silver nitrate	white ppt.
bromide, Br^- [in solution]	acidify with dilute nitric acid, then add aqueous silver nitrate	cream ppt.
iodide, I^- [in solution]	acidify with dilute nitric acid, then add aqueous silver nitrate	yellow ppt.
nitrate, NO_3^- [in solution]	add aqueous sodium hydroxide, then aluminium foil; warm carefully	ammonia produced
sulfate, SO_4^{2-} [in solution]	acidify with dilute nitric acid, then add aqueous barium nitrate	white ppt.
sulfite, SO_3^{2-}	add a small volume of acidified aqueous potassium manganate(VII)	the acidified aqueous potassium manganate(VII) changes colour from purple to colourless

Tests for aqueous cations

cation	effect of aqueous sodium hydroxide	effect of aqueous ammonia
aluminium, Al^{3+}	white ppt., soluble in excess, giving a colourless solution	white ppt., insoluble in excess
ammonium, NH_4^+	ammonia produced on warming	—
calcium, Ca^{2+}	white ppt., insoluble in excess	no ppt. or very slight white ppt.
chromium(III), Cr^{3+}	green ppt., soluble in excess	green ppt., insoluble in excess
copper(II), Cu^{2+}	light blue ppt., insoluble in excess	light blue ppt., soluble in excess, giving a dark blue solution
iron(II), Fe^{2+}	green ppt., insoluble in excess, ppt. turns brown near surface on standing	green ppt., insoluble in excess, ppt. turns brown near surface on standing
iron(III), Fe^{3+}	red-brown ppt., insoluble in excess	red-brown ppt., insoluble in excess
zinc, Zn^{2+}	white ppt., soluble in excess, giving a colourless solution	white ppt., soluble in excess, giving a colourless solution

Tests for gases

gas	test and test result
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	turns limewater milky
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	'pops' with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns acidified aqueous potassium manganate(VII) from purple to colourless

Flame tests for metal ions

metal ion	flame colour
lithium, Li^+	red
sodium, Na^+	yellow
potassium, K^+	lilac
calcium, Ca^{2+}	orange-red
barium, Ba^{2+}	light green
copper(II), Cu^{2+}	blue-green

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